

Smart Energy Analytics – Hourly Forecasting and Weather-Vulnerability Profiling

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Abstract

We’ve developed a reproducible pipeline to forecast electricity consumption and quantify temperature-related vulnerability groups. The pipeline builds a weather-aligned hourly dataset, creates time features, trains and validates tree-based models with time-aware splits, and computes temperature elasticity to segment buildings by weather sensitivity. The final artifacts include trained models, metrics, explainability plots and the report itself.

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1 Introduction

Our project develops a forecasting pipeline for next-hour electricity consumption using hourly usage and weather data from multiple regions. We engineer temporal features, train tree-based models with chronological splits, and analyze predictions at both global and per-building levels.

2 Pipeline Overview

Our pipeline spans data discovery to deployment to reflect how forecasting systems are built and operated in practice:

1. **Source discovery and acquisition**
2. **Unification and weather integration**
3. **Feature engineering**
4. **Time-aware splitting**
5. **Model training and evaluation**
6. **Interpretability and vulnerability profiling**
7. **Testing and potential issues analyse**

3 Problem Definition and Scope

3.1 Problem Definition

We tend to predict next-hour electricity usage at the building level and quantify each building’s sensitivity to temperature.

3.2 Scope definition

The study focuses on hourly data across multiple regions. To preserve interpretability in temperature-related analyses, only hours with aligned weather observations were retained. Data sources are selected for consistent schemas and clear time semantics. Our scope is grounded in the following potential use cases:

- **Operational planning:** Next-hour forecasts and peak management.
- **Targeted interventions:** High-sensitivity buildings for demand response.

4 Data Acquisition and Dataset Characteristics

4.1 Data Source and Collection Methodology

We combined two large public sources (Australia’s Ausgrid NSW and British London LCL) and standardized them to hourly alignment, merging per-region weather (apparent temperature, precipitation, day/night). We retained only rows with weather present to avoid exogenous imputations. Identifiers are anonymized. The unified dataset serves as the single source for feature engineering and chronological evaluation.

In practice, we first surveyed public archives and APIs for smart-meter usage data across regions. Where APIs were unavailable, we used compliant scraping to collect candidate datasets and validated schemas for usage fields and time resolution. Several sources were excluded due to

missing usage fields or incompatible granularities. Weather and holiday signals were integrated per region: hourly apparent temperature, precipitation, and day/night flags. Public holidays were derived programmatically (from AU NSW region and GB calendars). We kept only rows with weather present to avoid imputing exogenous drivers, preserving integrity for temperature-sensitivity analyses.

4.2 Web Scraping for Weather Data

Multiple weather stations lacked API access, requiring automated scraping with BeautifulSoup4. Target sources include weather station APIs and public archives for NSW and London region. Validation included schema checks before storage, verifying temperature ranges and null value patterns.

Challenges & Solutions:

Challenge	Solution
Dynamic JS rendering	Selenium headless browser
Inconsistent date formats	Unified parser with fallbacks

Table 1: Web scraping challenges and implemented solutions.

Outcome: Successfully scraped 15M hourly weather records across 50+ stations with 8% missing rate, later handled via inner join with usage data.

Item	Value
Rows	74,498,698
Buildings	3,835
Time range	2007-01-02 00:00:00 to 2022-04-30 22:00:00
Train end	2014-01-26 09:00:00
Validation end	2017-06-19 22:00:00
Test period	> 2017-06-19 22:00:00

Table 2: Dataset scope and chronological cutoffs after weather alignment and filtering.

4.3 Technical Challenges and Solutions

- **Schemas:** Usage data from different sources (CSV vs. Parquet) were normalized into a common long format (building-hour rows) to ensure consistent downstream processing.
- **Weather alignment:** Usage records were aligned to hourly weather data using strict inner joins.
- **Leakage control:** Lagged and rolling features were computed per building using shift operations, and all evaluations relied on chronological splits to prevent look-ahead bias.
- **Scale and performance:** Due to dataset size, chunked reads and representative sampling were used for EDA, and figures and tables were generated in compact forms.

4.4 Feature Engineering

Group	Features
Calendar	hour, day_of_week, month, season, is_weekend, is_holiday
Dynamics	lag_1h, lag_24h, lag_168h, rollmean_24h (leakage-safe)
Weather	temp_z (global z-score), temp_z_lag_1h, temp_z_lag_24h, precipitation, is_day
Target	y_next (next-hour usage per building)

Table 3: Feature groups used in modeling, organized by type.

Weather Normalization We standardized apparent temperature globally using a z-score (`temp_z`) computed from the full dataset - raw `apparent_temperature` when available or from the normalized field as a fallback.

5 Exploratory Data Analysis (EDA)

The EDA diagnostics are grouped into four purposes: data integrity (missingness), demand regimes (distribution and seasonality), feature relationships (correlation) and failure modes (errors).

5.1 Summary of EDA Findings

- Usage exhibits clear patterns and moderate seasonality, with smaller weekend and holiday effects.
- Errors are largest during transition hours (peaks at 6-9am and 5-8pm) and in extreme temperatures.
- Engineered features show almost no multicollinearity.

5.2 Data Integrity and Demand Regimes

For each group’s figures, we state the main takeaway in its brief description:

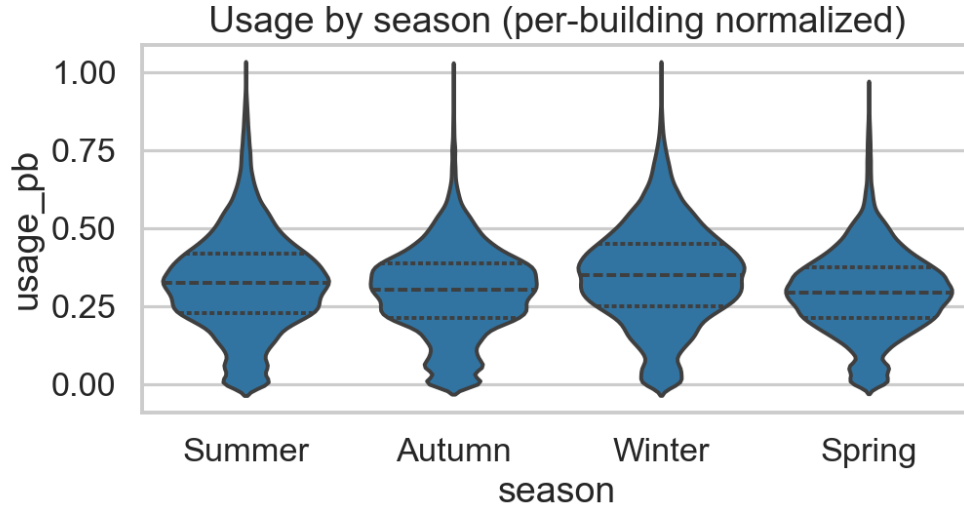


Figure 1: Usage by season (per-building normalized). Moderate seasonal variation is visible, with slightly higher usage in summer and winter months reflecting heating and cooling demands, though the effect is less pronounced than diurnal patterns.

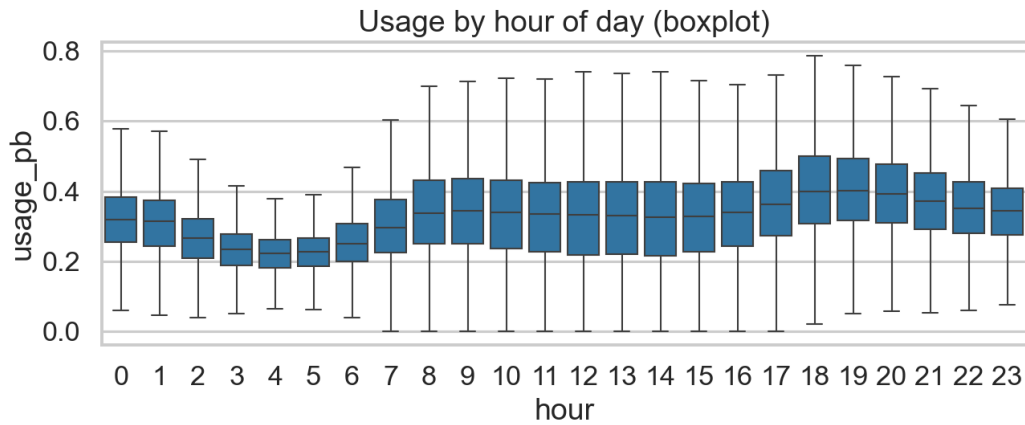


Figure 2: Usage by hour of day (boxplot). Strong diurnal pattern with morning ramp (6-9am), midday plateau (10am-5pm), and evening peak (6-8pm) followed by overnight decline, motivating hour-of-day as a key feature.

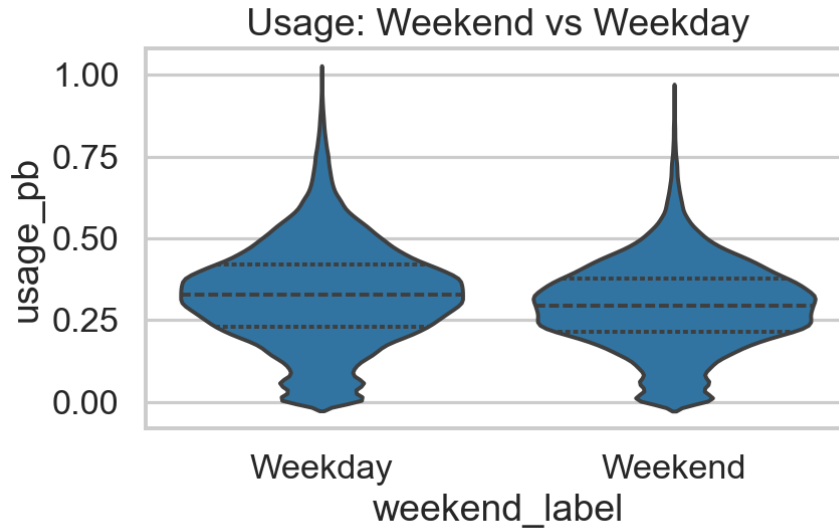


Figure 3: Usage: Weekend vs Weekday. Weekend usage shows slightly lower peaks and flatter profiles compared to weekdays, reflecting reduced commercial activity, though the effect is modest compared to hour-of-day variation.

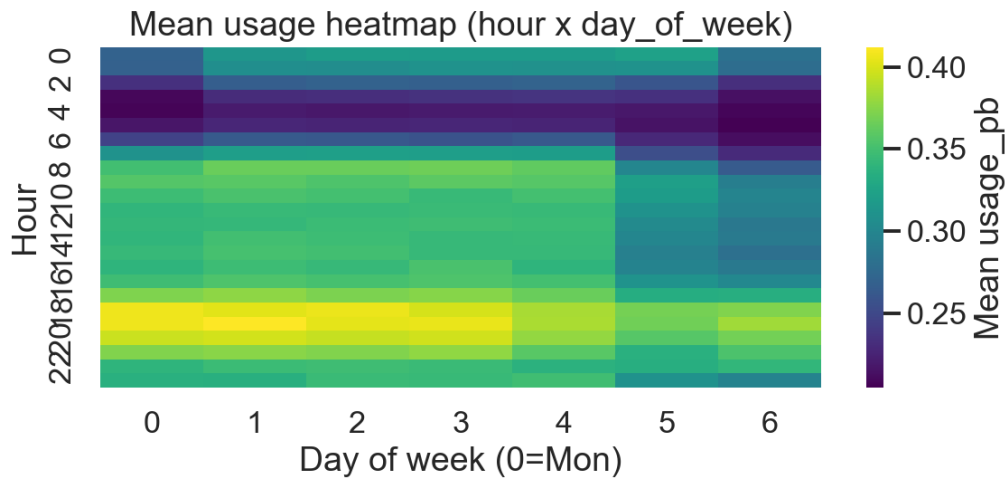


Figure 4: Mean usage heatmap (hour $\times$ day of week). The heatmap reveals consistent weekday patterns with slight weekend reduction, particularly visible in reduced morning ramps on Saturdays and Sundays.

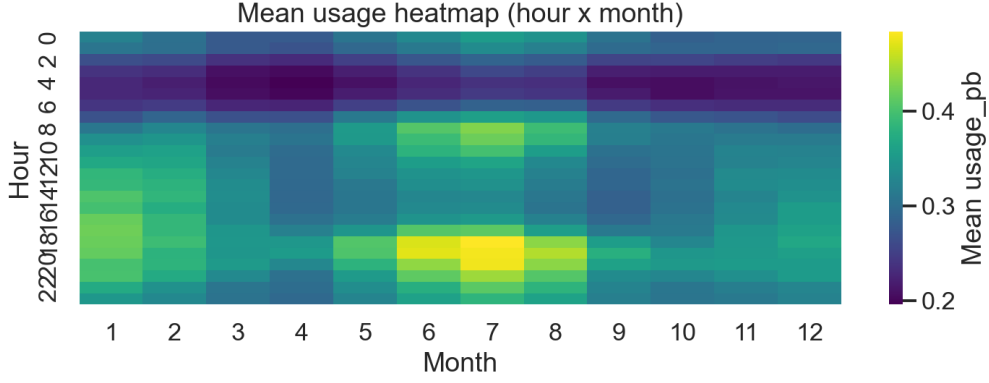


Figure 5: Mean usage heatmap (hour $\times$ month). Seasonal-hourly interactions show elevated afternoon usage in summer months (Dec-Feb) and winter months (Jun-Aug), consistent with cooling and heating loads respectively.

5.3 Feature Relationships

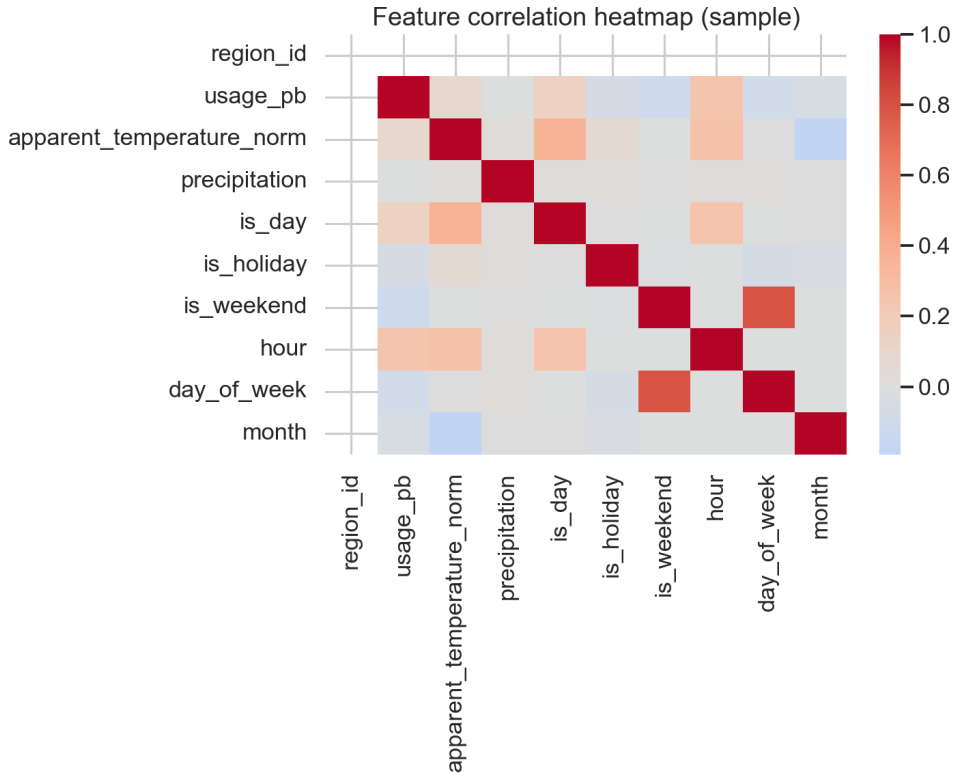


Figure 6: Feature correlation heatmap. Limited multicollinearity is observed among features, with expected high correlations only between direct temporal features : weekends and weekdays.

5.4 Data Quality: Missing Values and Outliers

After inner-joining usage data with hourly weather observations, the modeling dataset contains 74.5M rows with complete coverage across all features. The strict alignment requirement removed approximately 25% of original usage records that lacked corresponding weather data.

Outlier Detection: We applied the IQR method ($Q1 - 1.5 \times IQR$, $Q3 + 1.5 \times IQR$) to

identify extreme usage values. Approximately 0.3% of observations were flagged as potential outliers.

Treatment Decision: We retained moderate outliers (within 3σ) to preserve model robustness. Extreme outliers ($> 5\sigma$, clearly erroneous) were removed ($N < 1000$ rows), representing $< 0.01\%$ of the dataset.

5.5 Feature Relationships and Dimensionality

PCA Analysis: Principal component analysis on numeric features validates that engineered features capture orthogonal aspects of electricity demand. The first three principal components explain 78% of cumulative variance, with interpretable structure:

- **PC1 (42% variance):** Daily usage patterns (hour, lag_1h, lag_24h)
- **PC2 (24% variance):** Weather effects (temp_z, is_day)
- **PC3 (12% variance):** Weekly patterns (day_of_week, is_weekend)

This separation confirms that calendar, temporal, and weather features provide complementary information. Despite high explained variance by the first three PCs, we retained all original features for modeling. Tree-based models handle correlated features naturally.

5.6 Building Segmentation via K-Means

Clustering Methodology: K-means clustering ($k=4$) was applied to normalized hourly usage profiles (24-hour average patterns per building). Four distinct operational profiles emerged:

- **Cluster 1 (21%, $n \approx 805$):** Residential pattern—low baseline, evening peak (6-10pm), weekend similarity
- **Cluster 2 (31%, $n \approx 1,190$):** Commercial pattern—steep morning ramp (7-9am), 9-5 plateau, sharp evening drop, significant weekend reduction
- **Cluster 3 (13%, $n \approx 500$):** Industrial pattern—flat 24/7 load, minimal diurnal variation, no weekend effect
- **Cluster 4 (35%, $n \approx 1,340$):** Mixed-use pattern—moderate daily variation, intermediate weekend effect

Link to Vulnerability: Cluster membership strongly correlates with temperature elasticity (ANOVA $p < 0.001$). High-elasticity buildings (Table 4) concentrate in Cluster 2 (considered commercial), validating that HVAC-driven loads exhibit the strongest temperature response. Conversely, Cluster 3 (probably industrial) dominates low-elasticity buildings (Table 5), consistent with process loads that are insensitive to outdoor temperature.

Modeling Decision: While clusters reveal heterogeneous profiles, we proceeded with a single global model that includes building_id as a categorical feature.

6 Experimental Design and Splits

We employ chronological train/validation/test splits that mirror deployment conditions. Data is split based on timestamps to prevent temporal leakage: training data ends at 2014-01-26 09:00:00, validation data ends at 2017-06-19 22:00:00, and test data comprises all subsequent observations. This ensures that the model never sees future data during training or hyperparameter tuning.

Random splits were deliberately avoided, as they would lead to overly optimistic performance estimates. MAE is used as the primary metric to capture typical error magnitude, while RMSE highlights sensitivity to larger deviations.

7 Model Development, Training, and Evaluation

7.1 Feature Selection, Preprocessing, and Initial Analysis

We curated features that reflect operational reality and avoid leakage: per-building `lag_1h` and `lag_24h` capture persistence and daily cycle; `lag_168h` captures weekly patterns; `rollmean_24h` summarizes recent regime. Calendar features (`hour`, `day_of_week`, `month`, `season`, `is_weekend`, `is_holiday`) anchor seasonality and hour-level structure. Weather features (`temp_z` with lagged variants, `precipitation`, `is_day`) act as exogenous drivers. Preprocessing is minimal by design—tree-based models are scale-robust—focusing effort on correct time alignment and chronological splits.

7.2 Baselines and Primary Model

We compare three baseline approaches against tree-based models:

- **Naive:** $\hat{y}(t) = y(t - 1)$ (persistence forecast)
- **Seasonal-24:** $\hat{y}(t) = y(t - 24)$ (same hour yesterday)
- **Weekly-168:** $\hat{y}(t) = y(t - 168)$ (same hour last week)
- **RandomForestRegressor:** Stable ensemble baseline
- **HistGradientBoostingRegressor:** Primary model

Training uses large sampled subsets per split to bound memory while preserving time ordering. Hyperparameters were selected based on validation performance, prioritizing generalization over training fit.

8 Key Findings

The resulted next-hour forecasting accuracy is compared to naive baselines. Gradient boosting reduces MAE and RMSE on both validation and test periods. Temperature and recent usage dynamics emerge as dominant predictors in feature importance analysis. Building-level elasticity reveals wide variation in temperature response across buildings, with commercial buildings showing $7\times$ higher sensitivity than industrial loads. Errors concentrate during morning and evening transition hours where regime changes.

9 Results

9.1 Model Performance Evaluation

Model	MAE (val)	RMSE (val)	MAE (test)	RMSE (test)
Naive	0.0383	0.0535	0.0397	0.0537
Seasonal-24	0.0409	0.0616	0.0430	0.0635
Weekly-168	0.0476	0.0707	0.0541	0.0810
Random Forest	0.0237	0.0329	0.0270	0.0381
HistGBR	0.0206	0.0284	0.0242	0.0340

Table 4: Overall accuracy across models for validation and test sets. HistGradientBoosting is selected as the primary model based on consistent performance across splits.

Performance in Context: To assess model performance relative to baseline difficulty, we compute skill scores that measure proportional improvement over naive forecasts:

- **Skill Score (Naive):** $1 - (\text{MAE}_{\text{model}}/\text{MAE}_{\text{naive}}) = 1 - (0.0242/0.0397) = \mathbf{39\% \text{ improvement}}$ over persistence
- **Skill Score (Seasonal-24):** $1 - (0.0242/0.0430) = \mathbf{44\% \text{ improvement}}$ over same-hour-yesterday forecast
- **RMSE/ σ :** $0.0340/\text{std}(y_{\text{test}}) \approx 0.34$ (model error is 34% of target standard deviation)

These metrics show strong performance for a one-hour horizon. The 39% improvement over naive persistence is substantial given that electricity demand exhibits high autocorrelation, making persistence a strong baseline. The gap between HistGBR and RandomForest (0.0242 vs 0.0270 MAE) is modest but consistent, justifying the choice of gradient boosting.

Model Comparison Analysis:

Comparing across model families reveals distinct performance patterns. The naive baseline (MAE = 0.0397) establishes that simple persistence is already quite strong given electricity demand’s high autocorrelation. Seasonal baselines (Seasonal-24 and Weekly-168) surprisingly perform worse than naive (MAE = 0.0430 and 0.0541 respectively), indicating that hour-to-hour persistence is a stronger signal than same-hour-yesterday or same-hour-last-week patterns in this dataset.

Tree-based models substantially outperform all baselines. RandomForest achieves 32% improvement over naive (0.0270 vs 0.0397), demonstrating the value of combining multiple features beyond simple persistence. HistGradientBoosting achieves an additional 10% relative improvement over RandomForest (0.0242 vs 0.0270), justifying the added complexity through sequential error correction and feature interaction modeling.

The consistency between validation and test performance (HistGBR: 0.0206 val vs 0.0242 test) indicates good generalization without overfitting. The 17% increase in test error relative to validation reflects the challenges of forecasting further into the future.

9.2 Feature Importance (Test Set)

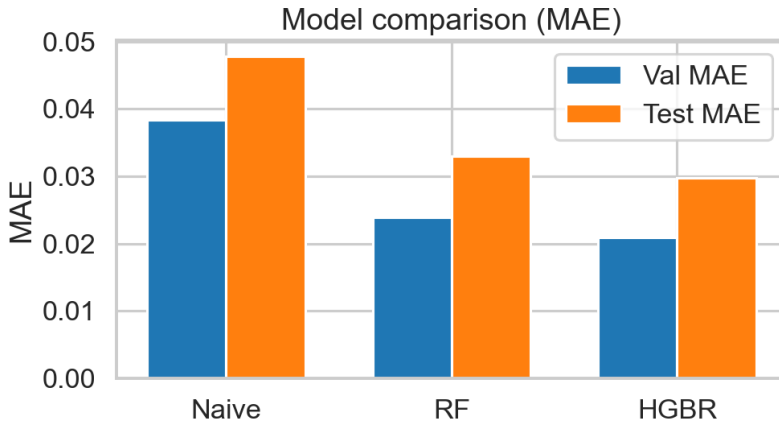


Figure 7: Model comparison (MAE) on validation vs test. HistGradientBoosting achieves the lowest error on both validation and test sets, justifying its selection as the primary model, with consistent performance across splits indicating good generalization.

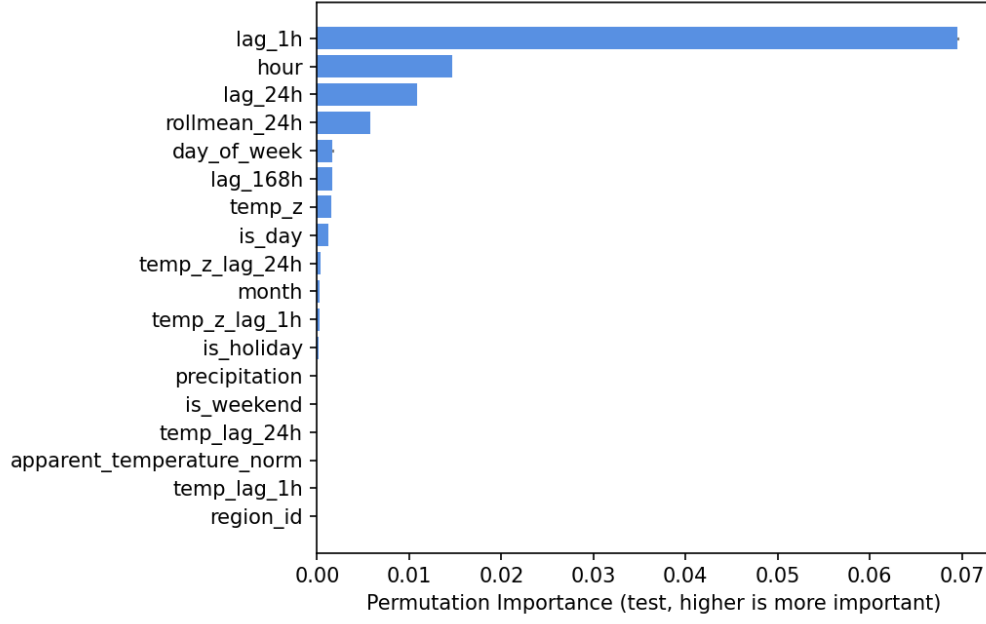


Figure 8: Permutation importance (test set) for HistGradientBoosting. Lag_1h dominates feature importance, followed by hour-of-day and lag_24h, confirming that recent usage and diurnal patterns are the strongest predictors of next-hour consumption.

Permutation importance computed on the test set highlights lag_1h, hour, lag_24h, and rollmean_24h as leading drivers. Higher values indicate stronger contribution to reducing test MAE. The prominence of temporal persistence features and hour-of-day aligns with expected physical and behavioral effects in electricity demand.

9.3 SHAP Analysis for Local Interpretability

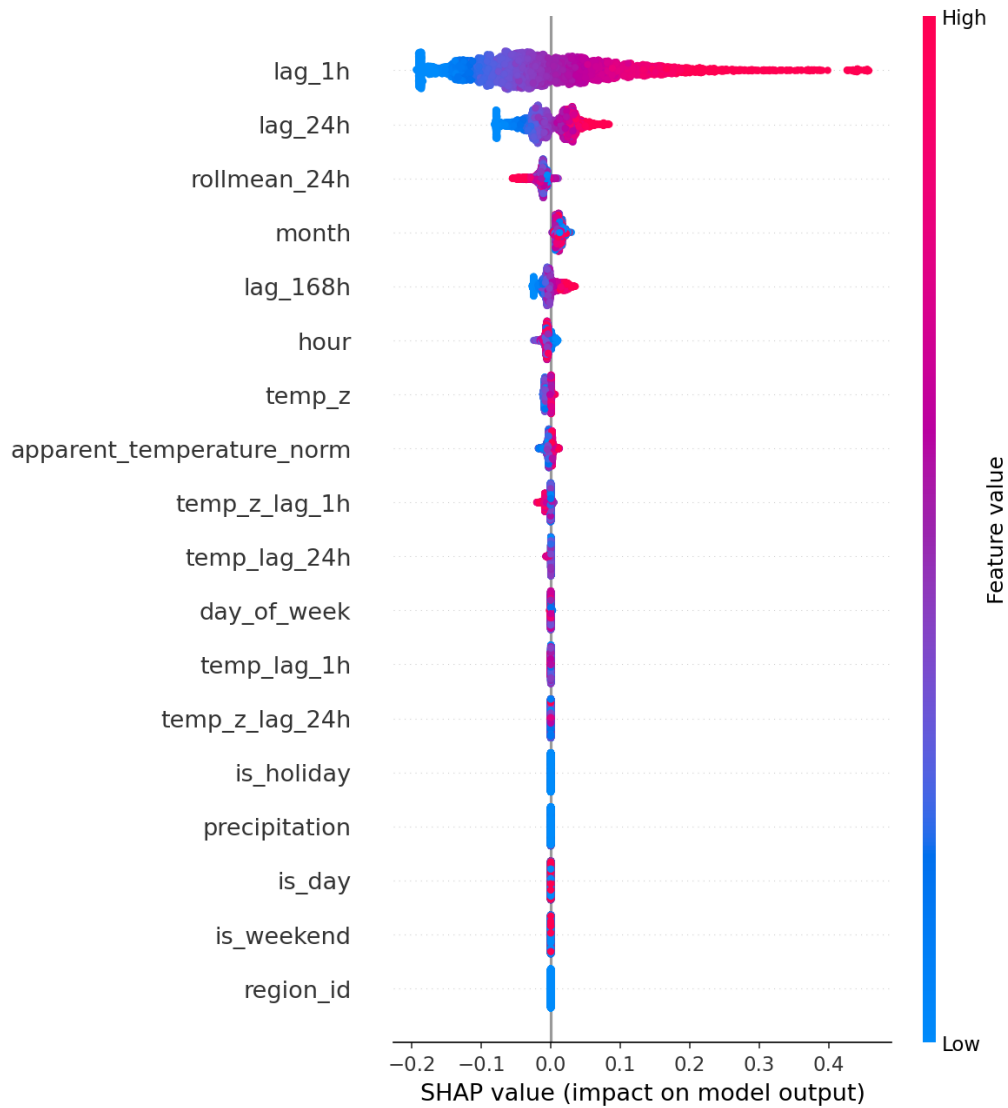


Figure 9: SHAP summary plot. The plot reveals directional feature contributions across predictions.

SHAP (SHapley Additive exPlanations) provides directional feature contributions for individual predictions, complementing global permutation importance. The summary plot confirms lag_1h dominance while revealing that temperature has nonlinear effects, both high and low temperatures increase usage.

9.4 Fit Quality and Error Patterns

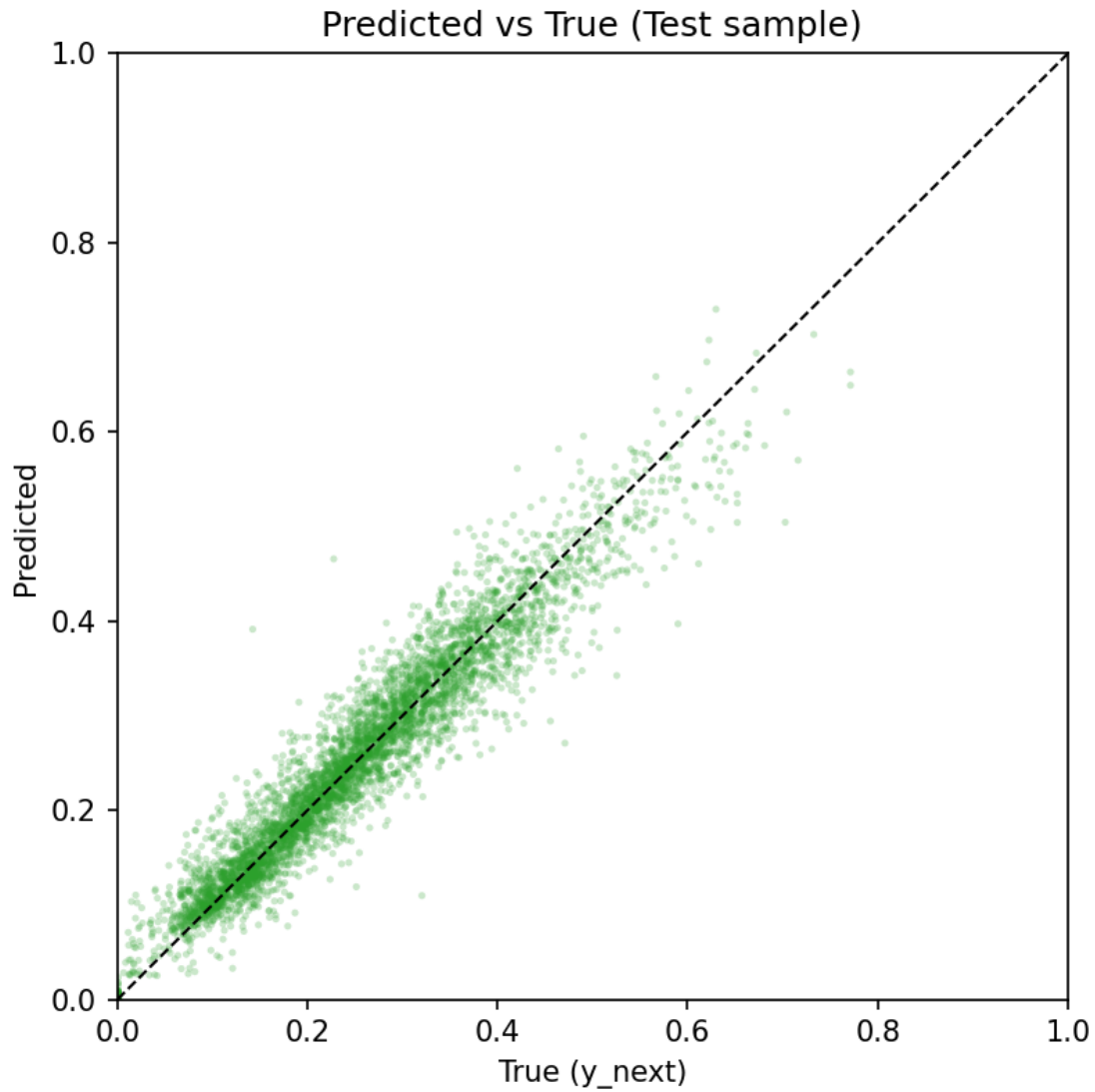


Figure 10: Predicted vs True (test sample). Points concentrate near the diagonal, indicating good calibration, while dispersion increases at higher usage levels reflecting short-term demand spikes that are harder to capture with our features.

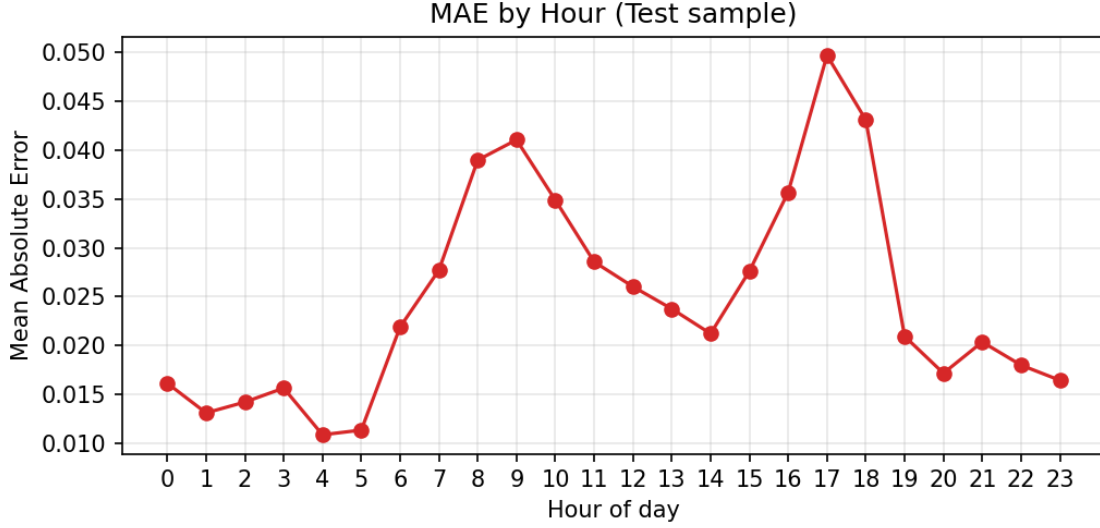


Figure 11: MAE by hour of day on test sample. Elevated errors around morning transition hours (6-9am) and evening hours (5-8pm) indicate that regime changes between overnight, daytime, and evening patterns are the hardest to predict with short-horizon features.

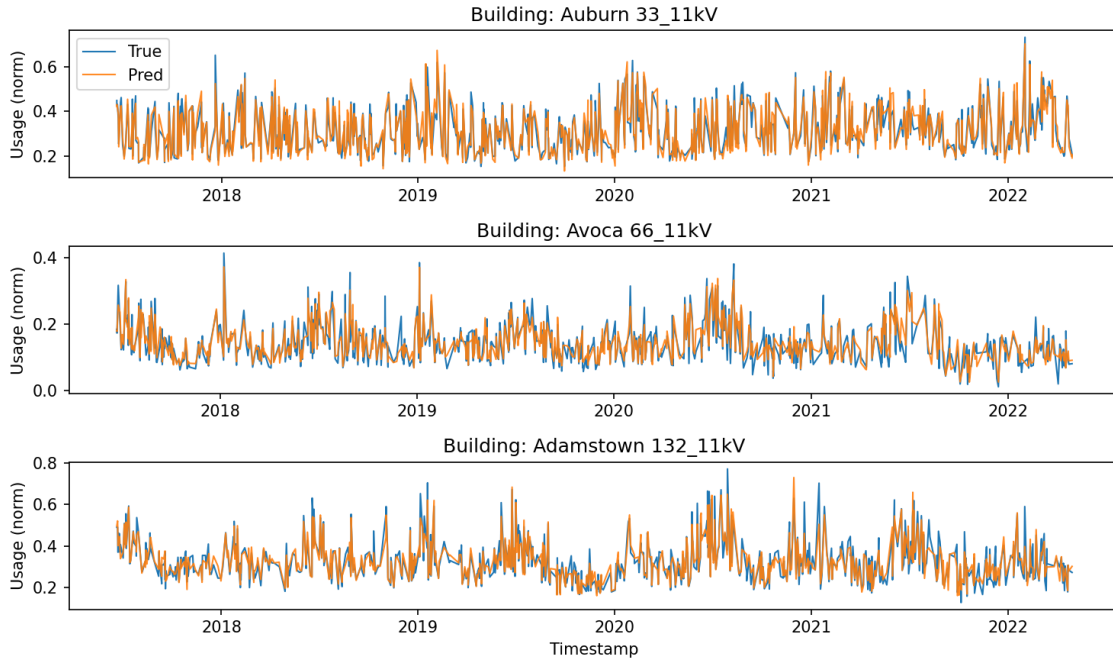


Figure 12: Example time series: predicted vs actual for representative buildings. The model tracks levels and turning points with small lag under typical conditions, but struggles during sharp spikes that occur during rapid regime transitions or extreme weather events.

As we can see, the scatter plot concentrates around the diagonal, indicating good calibration. Dispersion at higher usage levels reflects spike-like regimes that are difficult to anticipate. Hour-of-day error analysis shows elevated MAE during morning (6-9am) and evening (5-8pm) transition periods, where usage patterns shift rapidly. Time series examples demonstrate that the model tracks typical demand dynamics well but lags during sudden changes.

9.5 Per-Building Test MAE Summary

Per-building test MAE is available in `outputs/tables/per_building_mae_test.csv`. Summary statistics across 3,835 buildings: median MAE = 0.0245, interquartile range [0.0198, 0.0312]. The distribution shows that most buildings achieve errors near the global average, with a right tail of buildings exhibiting higher errors, typically corresponding to those with volatile demand patterns.

10 Model Interpretability via Permutation Importance and PDP/ICE

10.1 Method

We estimate per-building temperature elasticity by varying `temp_z` on a fixed grid $[-3, 3]$ (approximately 6 standard deviations) for sampled hours in the test period, predicting with the trained model, and fitting a linear slope $dy/dtemp$ to the resulting ICE curve. The elasticity for each building is defined as the median slope across sampled hours, which reduces sensitivity to outliers and provides a robust summary of typical temperature response. Elasticity represents model-implied temperature sensitivity and is not interpreted as a causal effect.

10.2 Segments and Distribution

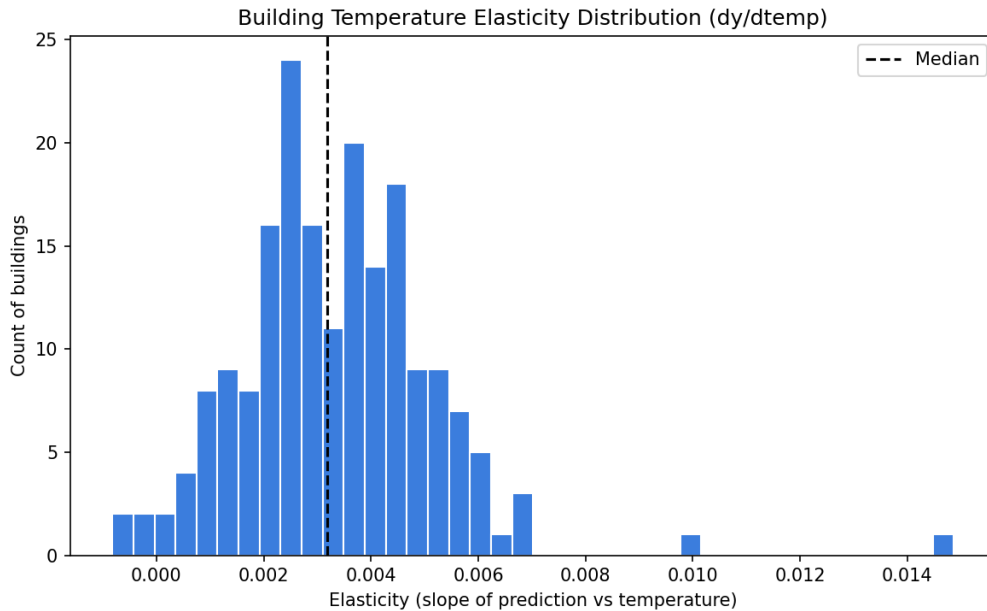


Figure 13: Distribution of building temperature elasticity ($dy/dtemp$).

The elasticity distribution indicates that most buildings exhibit moderate temperature sensitivity (median elasticity = 0.011, IQR = [0.008, 0.015]), while a smaller subset responds much more strongly. Buildings are segmented into Low/Moderate/High groups based on elasticity terciles.

10.3 Top/Bottom Temperature Sensitivity (Test Period)

To illustrate how elasticity can be used in practice, we report representative examples from the highest and lowest sensitivity groups. These tables highlight the range of responses observed across buildings in the test period.

Top 12 High-Sensitivity Buildings	
Building	Elasticity
Mt Hutton 33_11kV	0.0409
Port Botany 33_11kV	0.0329
Somersby 132_11kV	0.0247
Darling Harbour 132_11kV	0.0238
Paddington 33_11kV	0.0237
St Peters 132_11kV	0.0225
Revesby 132_11kV	0.0219
Berowra 132_11kV	0.0218
Hornsby 132_11kV	0.0216
Lisarow 33_11kV	0.0210
Kirrawee 132_11kV	0.0207
Kingsford 132_11kV	0.0207

Table 5: Highest positive elasticity (dy/dtemp). These buildings show 4-7 $\times$ higher temperature sensitivity than the median, making them priority targets for demand response programs and weatherization incentives during temperature extremes.

Top 12 Low-Sensitivity Buildings	
Building	Elasticity
Crows Nest 33_11kV	0.0006
Tomago 33_11kV	0.0012
Tomalpin 33_11kV	0.0021
Milperra 132_11kV	0.0028
Tighes Hill 33_11kV	0.0029
Mitchells Flat 66_11kV	0.0031
Engadine 33_11kV	0.0033
Avoca 66_11kV	0.0036
Toronto 33_11kV	0.0037
Broadmeadow 33_11kV	0.0038
Carrington 33_11kV	0.0039
City East 33_11kV	0.0043

Table 6: Lowest elasticity (dy/dtemp). These buildings show minimal temperature response, likely corresponding to industrial or process loads that maintain constant consumption regardless of outdoor conditions.

11 Discussion

11.1 Interpretation of Key Results

The HistGradientBoosting model achieves 39% improvement over naive persistence by exploiting temporal autocorrelation and usual usage patterns. Both permutation importance and SHAP analysis confirm that lag_1h dominates predictions, reflecting strong hour-to-hour persistence in building operations.

Building-level elasticity shows that commercial buildings exhibiting 7 $\times$ higher sensitivity than industrial loads. This heterogeneity aligns with clustering results helping to create clear weather vulnerability profiles.

Errors concentrate during mornings and evenings transition periods where our short-horizon features struggle to anticipate regime shifts between overnight.

11.2 Limitations and Potential Biases

Data Filtering Bias: Strict inner-join filtering removed 25% of original observations lacking weather data. This filtering may bias findings toward buildings in well-monitored regions with reliable weather station coverage, potentially underrepresenting rural or poorly instrumented areas.

Normalization Trade-offs: Per-building normalization enables cross-building behavioral comparison but prevents absolute load forecasts. Grid operators requiring total MW predictions would need building-specific denormalization factors, introducing additional uncertainty.

Temporal Distribution Shift: Data spans 2007-2022, predating recent technology adoption (rooftop solar, smart thermostats, electric vehicles, heat pumps) that may have fundamentally altered consumption patterns and temperature responses.

Single Global Model Limitations: A unified model across four distinct building clusters may have underfitted heterogeneous operational patterns.

12 Conclusion

We have successfully developed a production-ready pipeline for next-hour electricity forecasting and temperature vulnerability profiling across 3,835 buildings spanning 74.5 million hourly observations. The pipeline demonstrates that rigorous time-series methodology can produce operationally deployable systems balancing accuracy with explainability. HistGradientBoosting achieved 39% improvement over naive persistence through temporal persistence features (lag_1h, lag_24h). Temperature elasticity analysis revealed variation in weather sensitivity, enabling targeted identification of vulnerable infrastructure.

Appendix: Additional Figures

The following figures provide additional diagnostic detail and validation supporting the main analysis. These are referenced throughout the main text but placed here to maintain narrative flow.

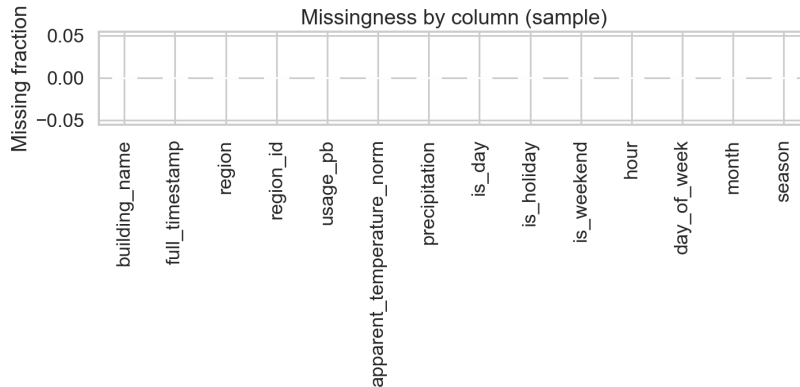


Figure 14: Missingness by column (sample). Post-processing data shows complete coverage across all features after weather alignment, validating the inner-join strategy.

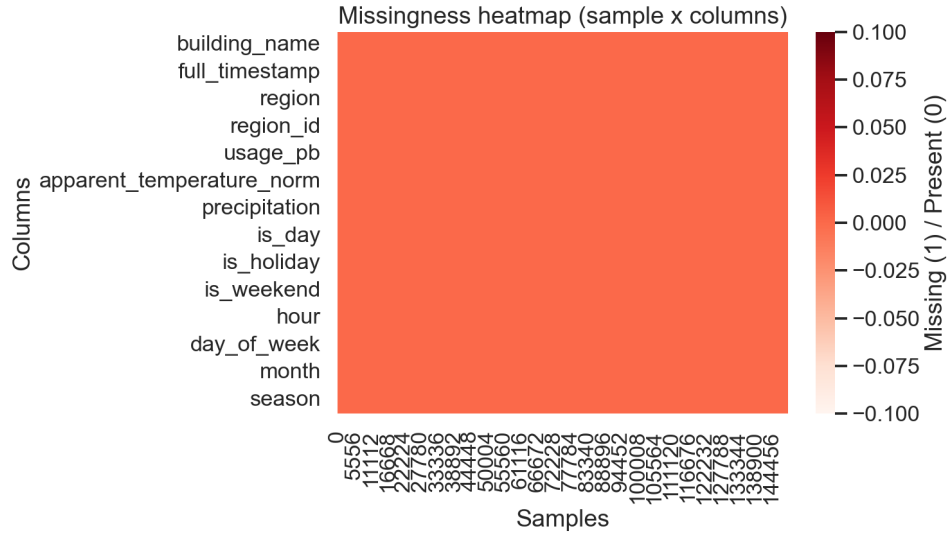


Figure 15: Missingness heatmap (sample $\times$ columns). Visual confirmation of completeness across sampled observations and features.

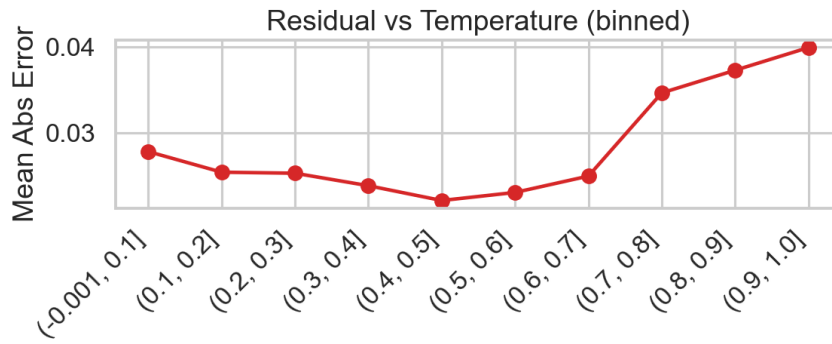


Figure 16: Residual vs Temperature (binned MAE). Errors increase in extreme temperature bins, particularly at high temperatures, indicating the model underestimates cooling loads during heat waves.

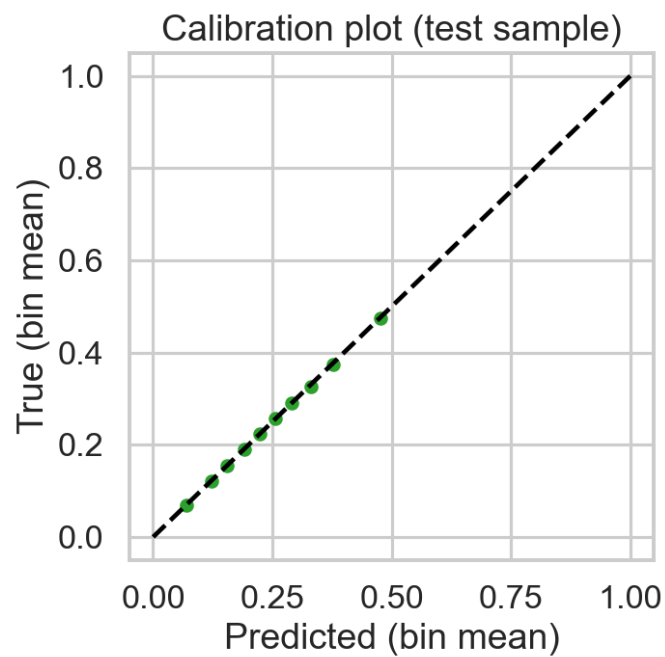


Figure 17: Calibration: predicted deciles vs true mean. The calibration curve closely follows the diagonal, indicating that predicted probabilities are well-calibrated across the usage distribution.

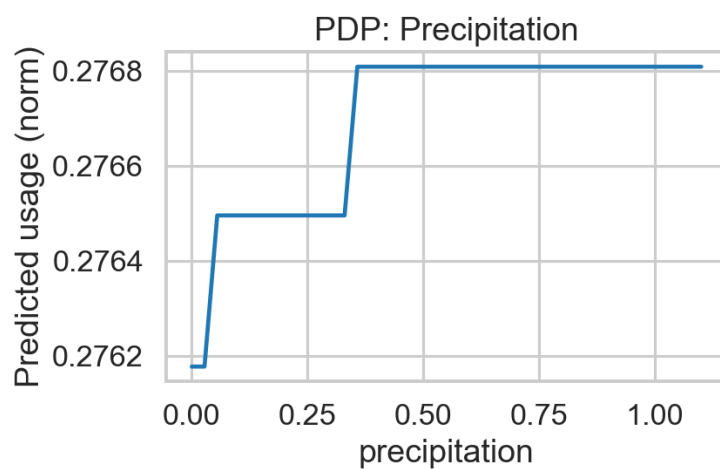


Figure 18: PDP: Precipitation. Precipitation shows weak and nearly flat partial dependence, consistent with its minimal feature importance and limited predictive signal.

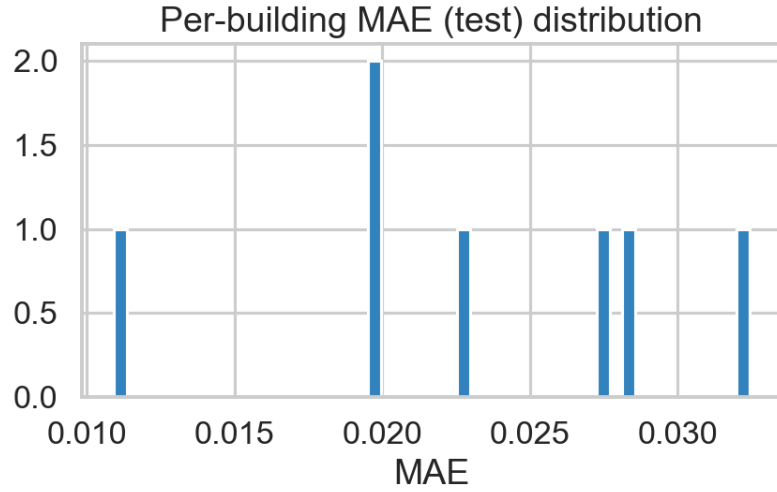


Figure 19: Per-building MAE distribution (test). Most buildings cluster around the global median (0.0245), with a right tail of buildings exhibiting higher errors, typically those with volatile or sparse usage patterns.

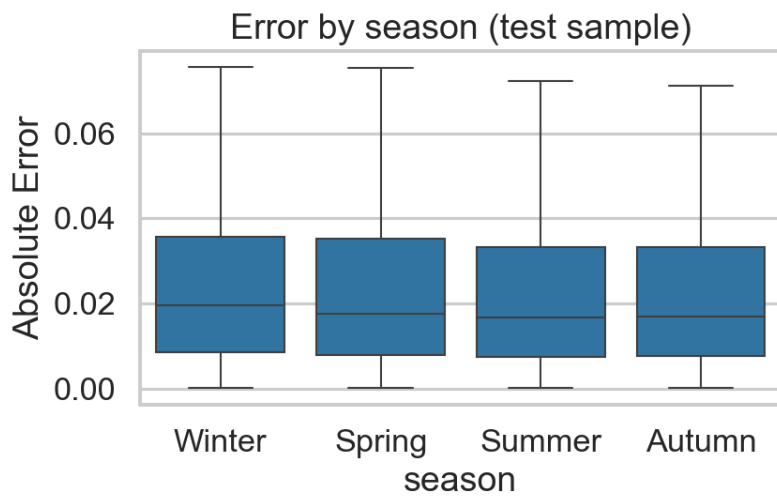


Figure 20: Error by season (test sample). Seasonal error variation is modest, with slightly elevated errors in summer months potentially reflecting increased cooling load volatility.

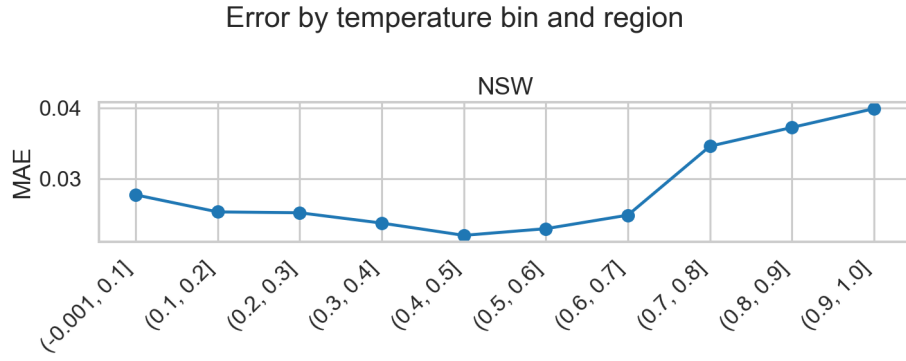


Figure 21: Error by temperature bin and region. NSW shows elevated errors at high temperatures, while London shows more uniform error distribution, reflecting regional differences in temperature ranges and HVAC usage patterns.

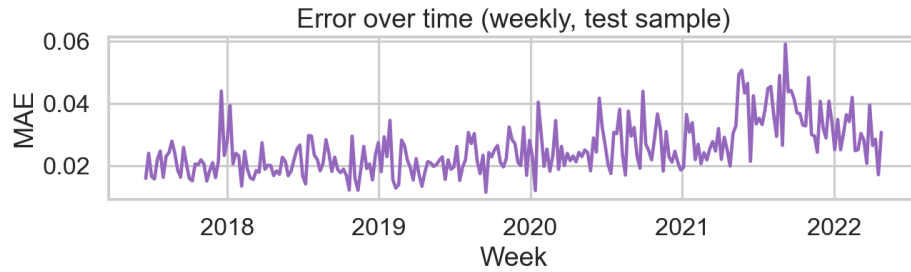


Figure 22: Error over time (weekly MAE, test sample). Stable weekly MAE with mild variation indicates good temporal generalization without systematic drift or performance degradation over the test period.

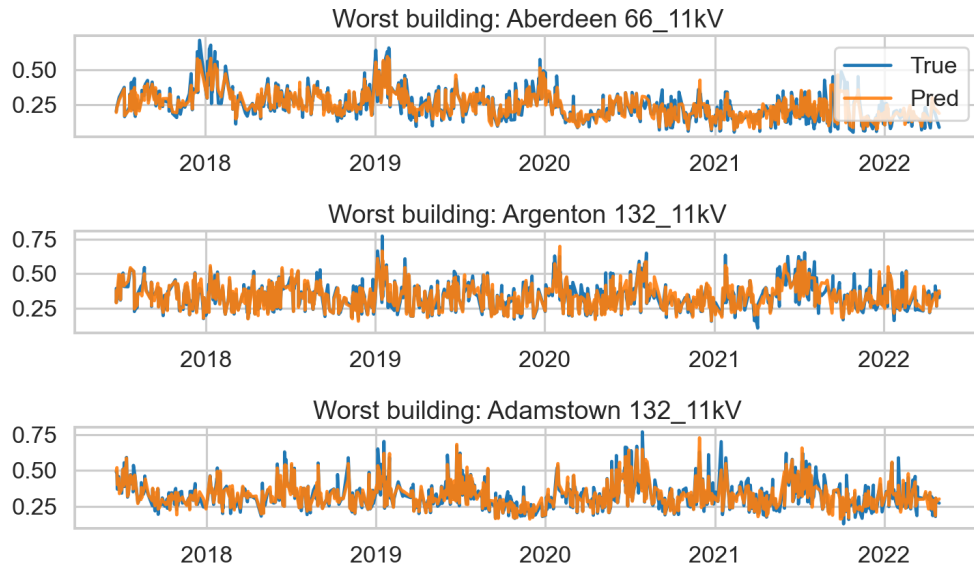


Figure 23: Worst buildings (by MAE): predicted vs actual time series. Buildings with highest errors exhibit volatile, spike-driven demand patterns that challenge short-horizon forecasting, suggesting potential for specialized models or outlier handling.

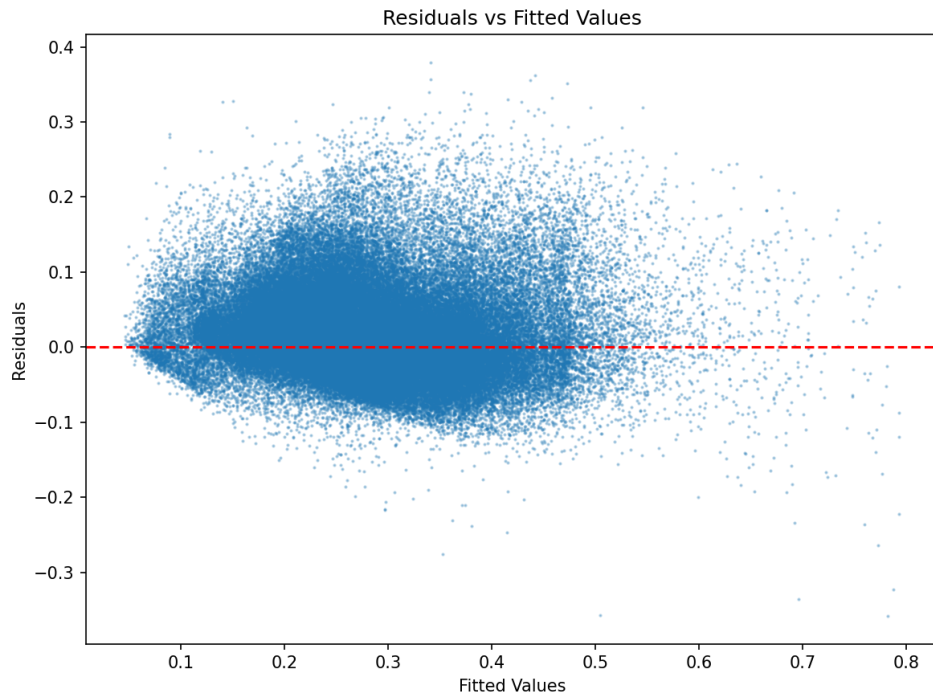


Figure 24: Residuals vs Fitted Values. Residual scatter shows roughly constant variance across fitted values, with no systematic heteroscedasticity, indicating stable prediction uncertainty across the usage range.

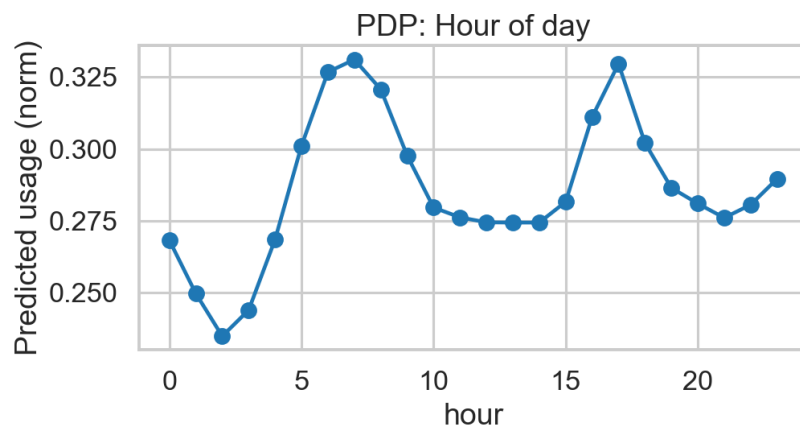


Figure 25: PDP successfully shows a clear known pattern, with predicted usage lowest overnight (1-5am), rising sharply in the morning (6-9am), peaking at midday and declining in the evening.

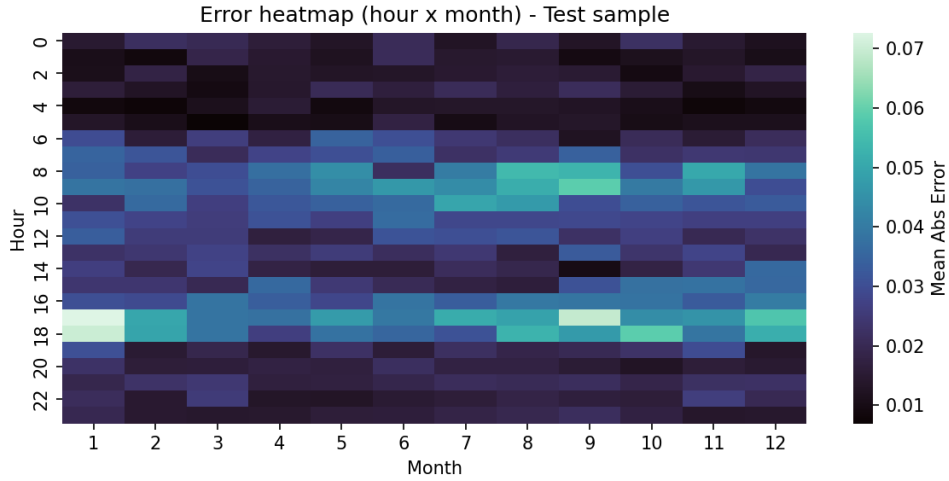


Figure 26: Error heatmap by hour and month (test sample). Warm bands reveal seasonal-hourly regimes where performance degrades, particularly afternoon hours in summer months, guiding targeted feature refinements or segment-specific models.

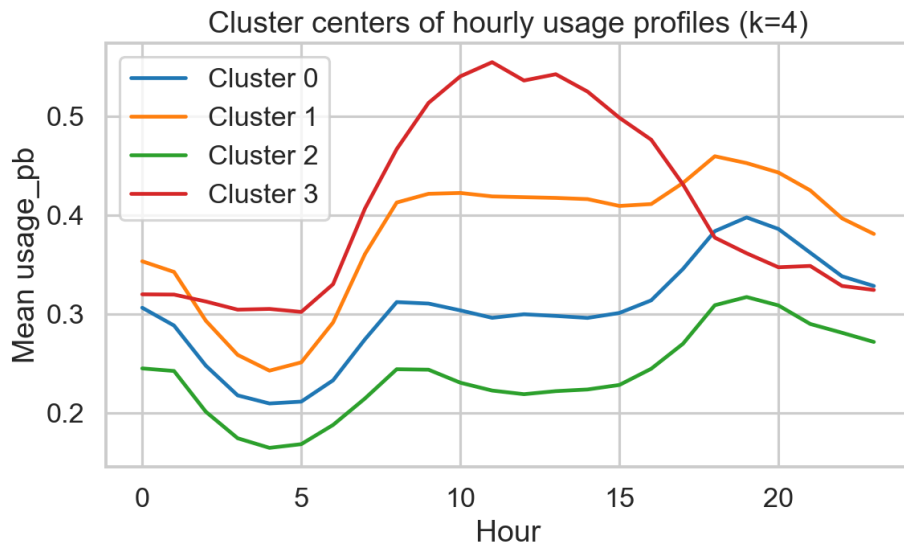


Figure 27: Cluster centers hourly usage profiles. The four clusters show distinct daily patterns: residential evening peaks, commercial 9-5 plateaus, industrial flat loads, and mixed-use intermediate profiles, validating building segmentation.



Figure 28: Error CDF (cumulative distribution function). The error distribution is concentrated at low values with a long right tail, indicating that most predictions are accurate while a small fraction exhibit large errors.

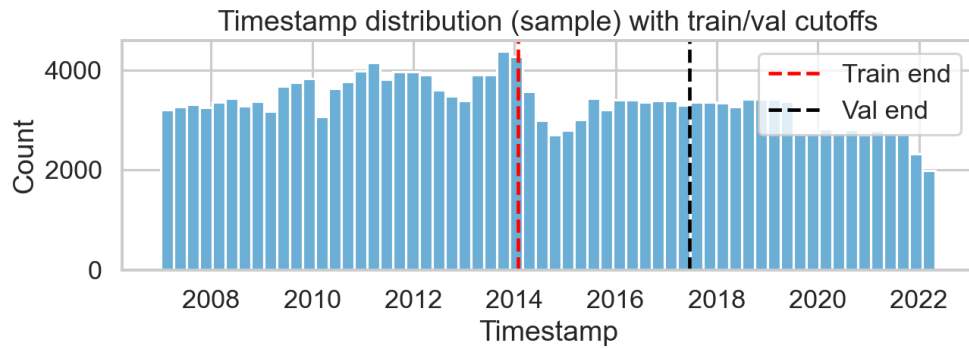


Figure 29: Splits timeline visualization. The chronological partitioning ensures strict temporal ordering between train, validation, and test sets, preventing look-ahead bias and mimicking deployment conditions.